A Network Model of HIV Transmission

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Introduction

- HIV is still a large, global problem

According to United Nations stats:

- A total of more than 80 million HIV cases since the beginning of the AIDS epidemic, resulting in more than 40 million deaths
- Over 40 million people still living with HIV in 2025
- Over 1 million new cases of HIV in 2025
- 570,000 estimated deaths from HIV related causes in 2025

Datasets: Transmission and Demographics

- Hladik W, Stupp P, McCracken SD, Justman J, Ndongmo C, Shang J, Dokubo EK, Gummerson E, Koui I, Bodika S, Lobognon R, Brou H, Ryan C, Brown K, Nuwagaba-Biribonwoha H, Kingwara L, Young P, Bronson M, Chege D, Malewo O, Mengistu Y, Koen F, Jahn A, Auld A, Jonnalagadda S, Radin E, Hamunime N, Williams DB, Kayirangwa E, Mugisha V, Mdodo R, Delgado S, Kirungi W, Nelson L, West C, Biraro S, Dzekedzeke K, Barradas D, Mugurungi O, Balachandra S, Kilmarx PH, Musuka G, Patel H, Parekh B, Sleeman K, Domaoal RA, Rutherford G, Motsoane T, Bissek AZ, Farahani M, Voetsch AC. The epidemiology of HIV population viral load in twelve sub-Saharan African countries. PLoS One. 2023 Jun 26;18(6):e0275560. doi: 10.1371/journal.pone.0275560. PMID: 37363921; PMCID: PMC10292693.
- Gedefie A, Muche A, Mohammed A, Ayres A, Melak D, Abeje ET, Bayou FD, Belege Getaneh F, Asmare L and Endawkie A (2025) Prevalence and determinants of HIV among reproductive-age women (15–49-years) in Africa from 2010 to 2019: a multilevel analysis of demographic and health survey data. Front. Public Health 12:1376235. doi: 10.3389/fpubh.2024.1376235
- UNAIDS. “Global HIV & AIDS Statistics — 2024 Fact Sheet.” UNAIDS, 2025, www.unaids.org/en/resources/fact-sheet.

Datasets: Mortality

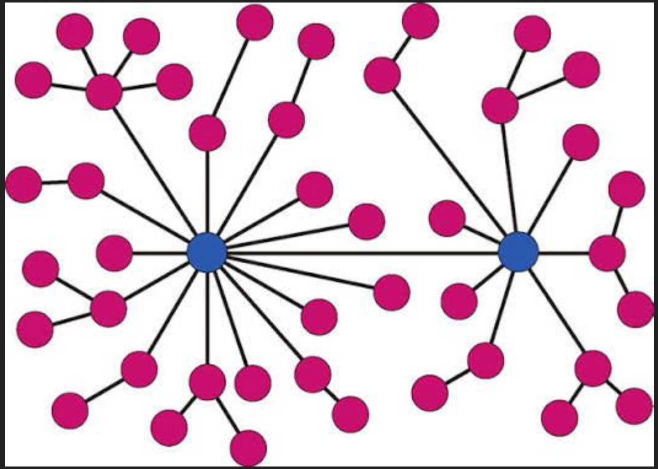
- Scheier, Thomas C.a,*; De Gouveia, Keishab,*; Engel, Mark E.c,d; Hohlfeld, Ameer S.-J.d; Cen, Alexe; Berhe, Annef; Fan, Sabrinae; Li, Jefferye; Elliott, Shakeape; Ford, Nathang,h; Meintjes, Graemei,j; Mertz, Dominika,k,l; Eikelboom, Johna,l; Wasserman, Seanj,m,n. One-year mortality among adults with advanced HIV in sub-Saharan Africa: a systematic review and meta-analysis. AIDS 40(7):p 967-981, July 2026. | DOI: 10.1097/QAD.0000000000004431.
- Nacarapa, E., Maddalozzo, R., Moosa, MY.S. et al. Baseline characteristics associated with mortality among children living with HIV initiating ART at a rural district HIV clinic of Mozambique. Sci Rep (2026). https://doi.org/10.1038/s41598-026-36433-1.

Datasets: Modelling

- Cooper, Ian, et al. “A SIR Model Assumption for the Spread of COVID-19 in Different Communities.” Chaos, Solitons & Fractals, vol. 139, Oct. 2020, p. 110057, https://doi.org/10.1016/j.chaos.2020.110057.
- Eames, K.T.D. “Modelling Disease Spread through Random and Regular Contacts in Clustered Populations.” Theoretical Population Biology, vol. 73, no. 1, Feb. 2008, pp. 104–111, https://doi.org/10.1016/j.tpb.2007.09.007. Accessed 7 May 2020.
- Rodrigo Volmir Anderle, et al. “Modelling HIV/AIDS Epidemiological Complexity: A Scoping Review of Agent-Based Models and Their Application.” PLOS ONE, vol. 19, no. 2, 2 Feb. 2024, pp. e0297247–e0297247, https://doi.org/10.1371/journal.pone.0297247. Accessed 2 Apr. 2024.
- Libin, Pieter & Hernalsteen, Laurens & Theys, Kristof & Gomes, Perpétua & Abecasis, Ana & Nowe, Ann. (2018). Bayesian inference of set-point viral load transmission models. 10.48550/arXiv.1811.11042.

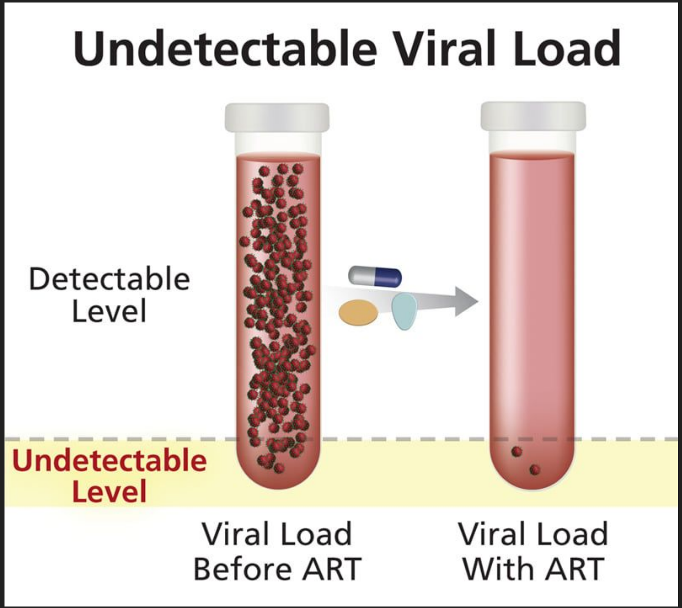
Methods: Graph

- Our model works by modelling individuals as nodes in a graph. We are using the package NetworkX as a base for lots of graph operations. This allows us to track connections between people, as well as how those connections chain through each other; A is connected to B, who is connected to C, etc.



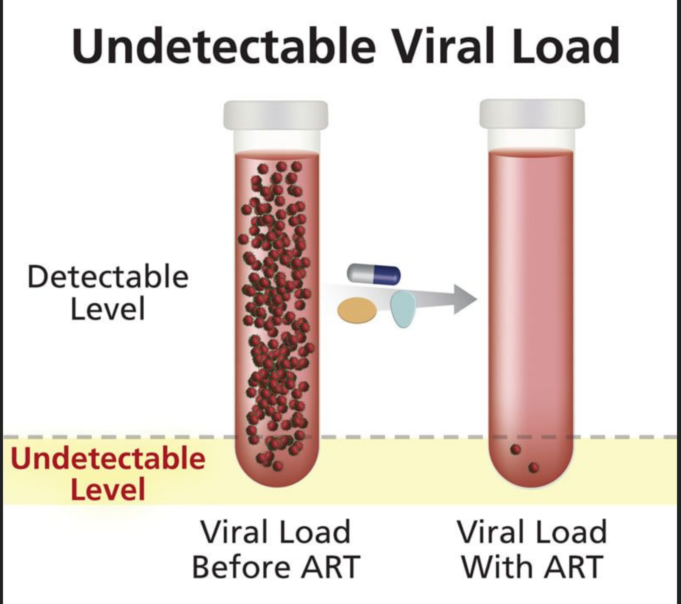
Methods: Viral Load Modelling

- HIV infection is more complex than binary. The viral load of an individual is the copies of the virus that can be expected to be found in a milliliter of blood; a higher viral load, or VL, means the infection is more liable to spread and poses a greater danger to the individual.



Methods: Viral Load Modelling

- By individually tracking each imaginary person's viral load, we can control more levers and bring the simulation closer to a useful reflection of reality. This somewhat replaces the SIR model, since we do not use binary infection. Instead, we have unsuppressed, suppressed, and undetectable.



Methods: Steps

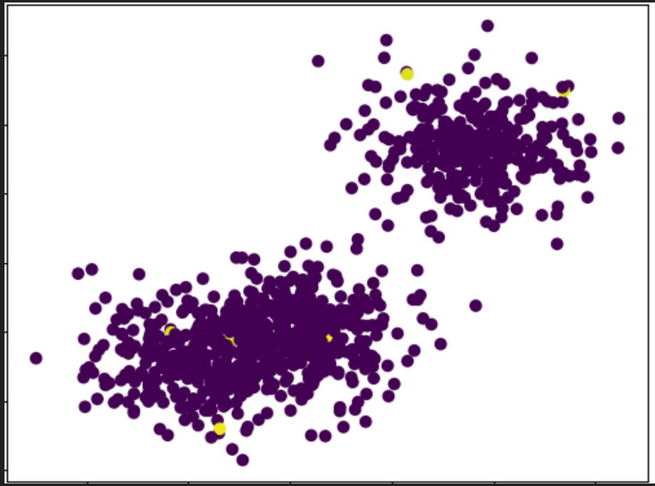
- We will set up our population according to some parameters, and then we will run a series of “ticks” in the simulation, each representing a week in real time.
- Each tick of the simulation will
  - calculate the infections that occur from each node to each of their neighbors,
  - calculate the change in viral load for all infected nodes
  - calculate the mortality rate of any nodes that have severe untreated viral levels, and then
  - calculate the progression of medical efforts to put some individuals on medical treatment for HIV called Anti-Retroviral Therapy, or ART.

Methods: Simulation Setup

- Setting up the simulation requires several parameters. There are two things we want to consider: geographical location and “close contacts”.
    - Geographical location will be used to represent a small likelihood of proximal spread, and
    - close contacts will be used to represent connections that represent a likely vector of spread for HIV.

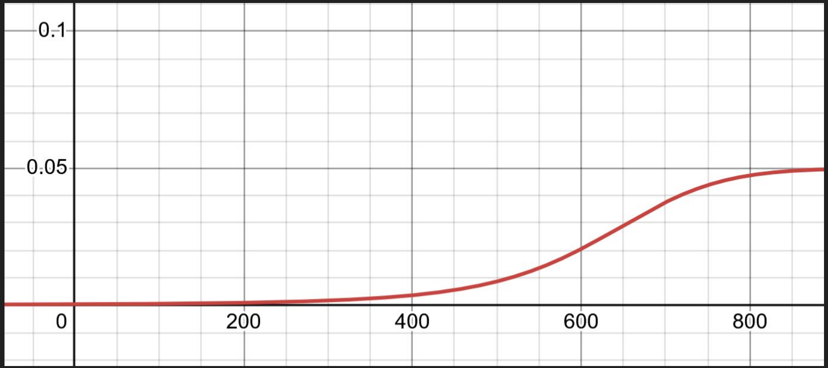
Methods: Simulation Setup

For geographical location, we generate some number of "cities" that the population on the graph will center around. These "city center locations" are placed around the map randomly, and then each individual node in the graph will be placed into one of these random cities, distributed some distance from the center, according to a user-specified distribution (like a t-distribution or normal distribution, depending on the preferred balance of urban vs. rural population). Then, some random nodes are chosen to be infected with a small initial viral load, according to a user-specified initial infection probability.



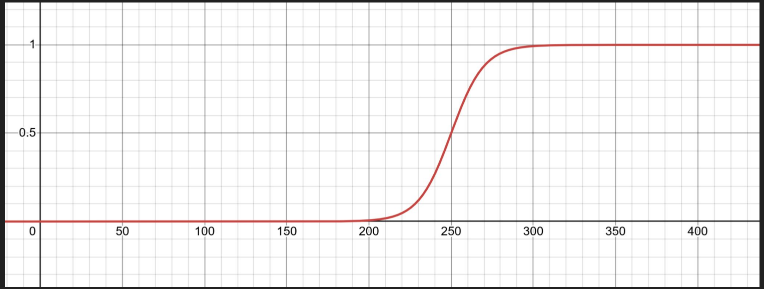
Methods: Mortality

- To tick the graph, we have several steps.
    - First we determine any deaths due to unsuppressed viral load. The way we do this by plugging in the viral load for an infected individual into a kind of chained exponential logistic function. This is done to mirror individual death rates: the year-to-year death rate for a completely unsuppressed viral load means that the death rate at a completely unchecked viral load should only be around 5% for a weekly tick.



Methods: Transmission

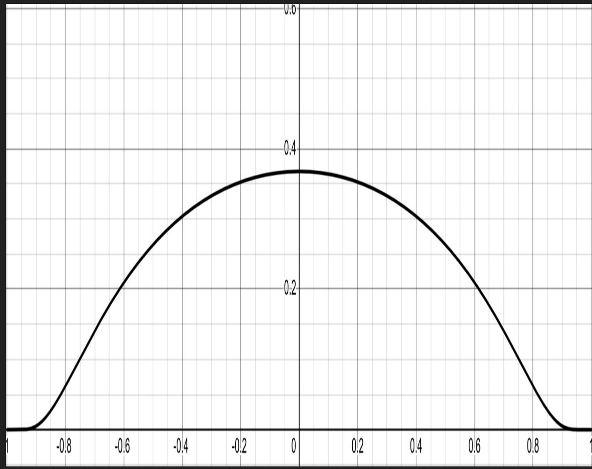
Then, each infected node will check its infection probability for each other node in the graph. Each node will check all nodes that it has a close connection edge with (which has no connection with geographic distance) and generate a probability based on the viral load of the infected individual. This viral load infection function is a logistic curve, with low infection probability for low viral load, nearing 0 at what is considered "undetectable" viral load levels of 50 or lower.



Methods: Proximal Transmission

Then another check is made for proximal, distance-based infection, with about a ~20% lower rate, varying depending on the distance between the two nodes.

- Old versions of the simulation had this check between every two nodes of the graph, meaning the runtime for each simulation would scale quadratically with the number of nodes in the graph. At some point, I implemented a caching system whereby the simulation just remembers the nodes that have a nonzero chance to have proximal spread between them, so it went back to scaling linearly at the cost of some overhead time and some additional space for storing proximal neighbors.

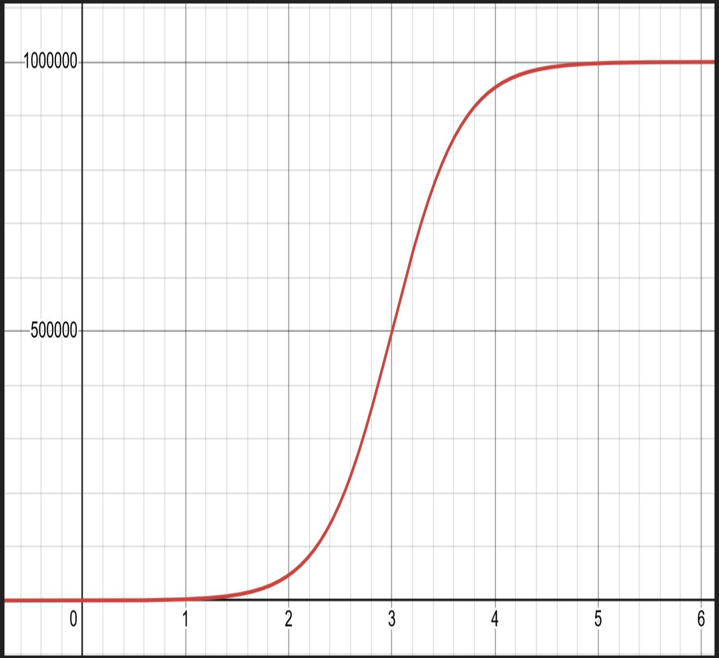


Methods: Suppressed Viral Load Progression

Then the simulation checks to change viral load based on if each individual is on ART or not, a characteristic which is stored for each node. For nodes on ART, this is done using an exponential decay function, where the viral load will very quickly drop if viral load is high, and will slowly settle at an undetectable level below 50 copies per mL.

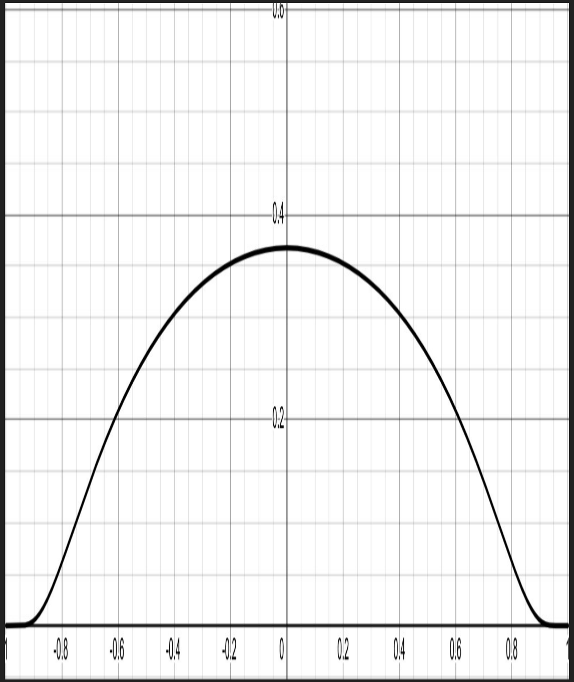
Methods: Viral Load Progression

For unsuppressed individuals without ART, this is done by using the derivative of a logistic function that models untreated viral load progression and modifying viral load by that amount. Viral load increases quickly and then reaches a maximum, so I modeled it using a logistic function.

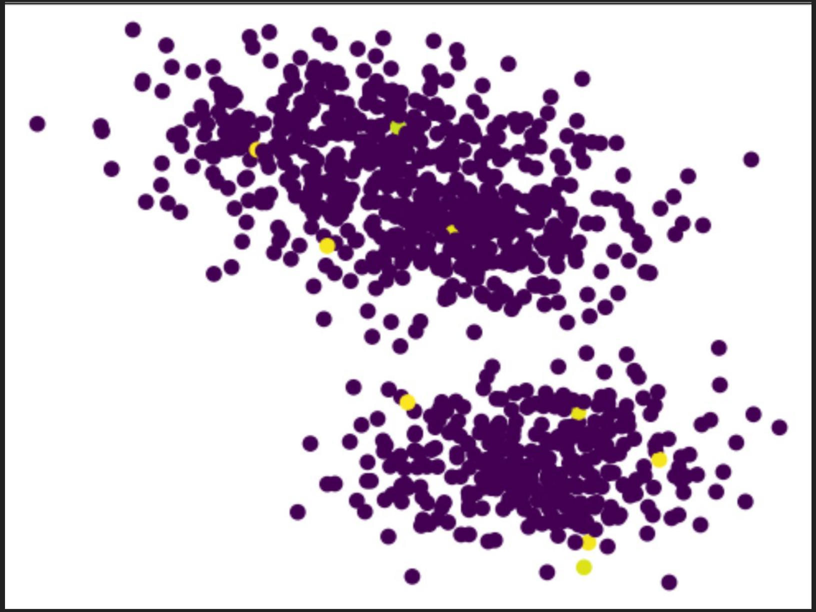


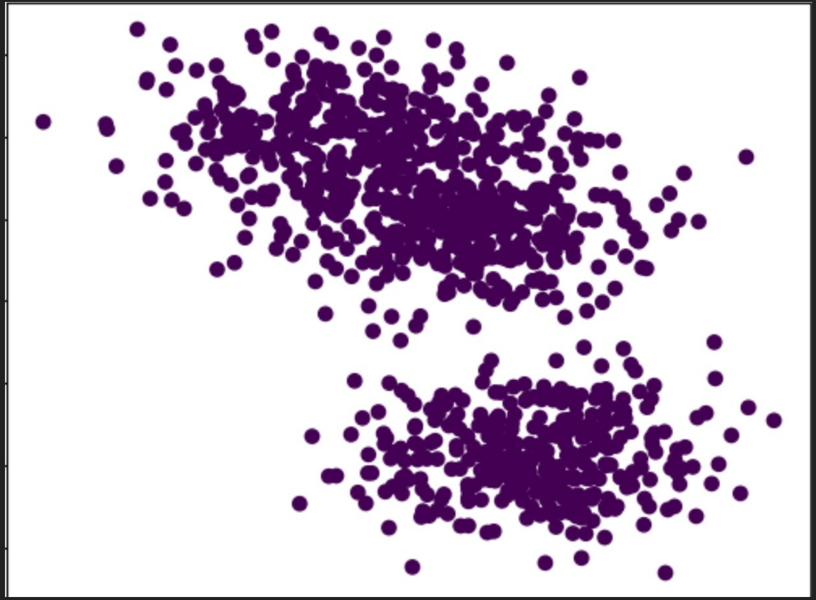
Methods: ART Initiation

Then the simulation checks to initiate some nodes for ART. During simulation setup, a specified number of clinics are placed in geographic locations following similar rules to individuals. Each tick, an infected node has a certain chance to initiate ART based on its distance to a clinic. This uses the same bump function as proximal infection chances, and can be tweaked to specify maximum chance of initiation as well as effective radius of these clinics.

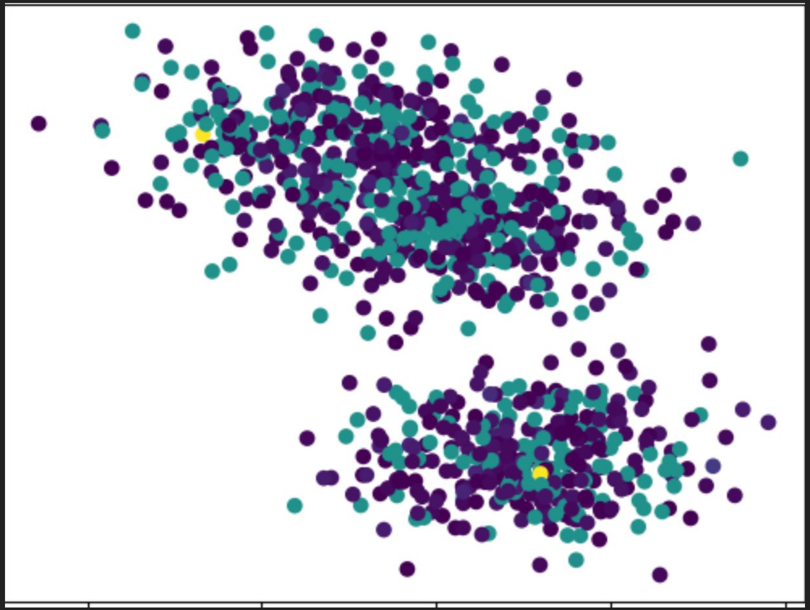


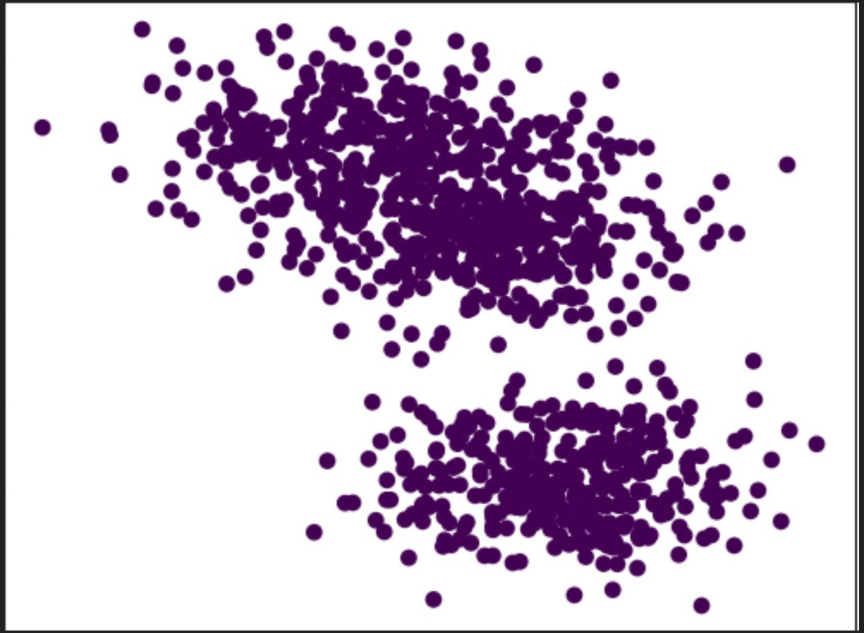
Results: Simulation Demonstration - Initial



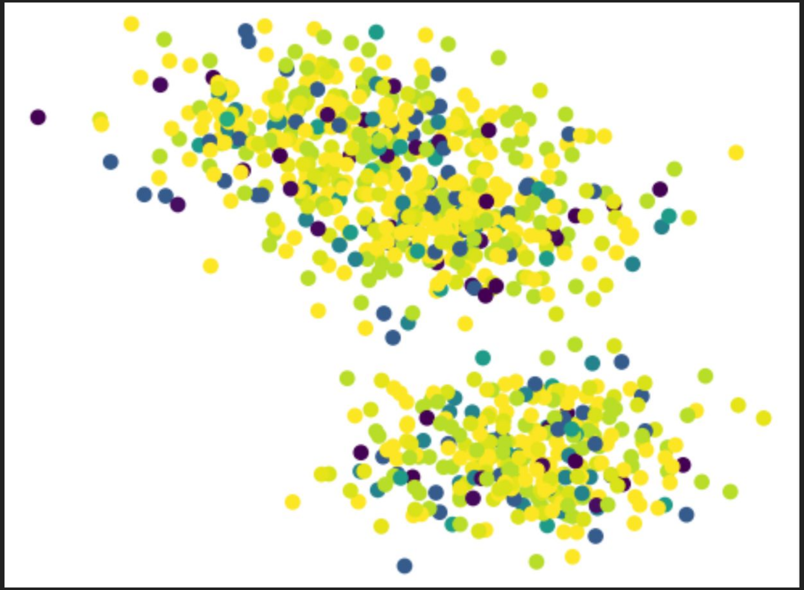


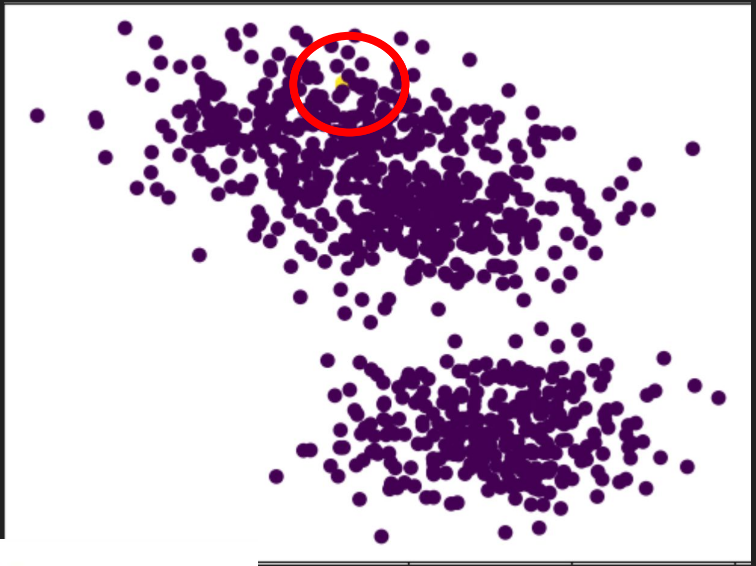
Results: Simulation Demonstration - 1 Week

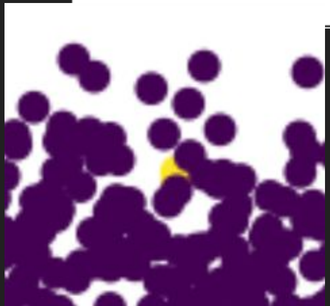




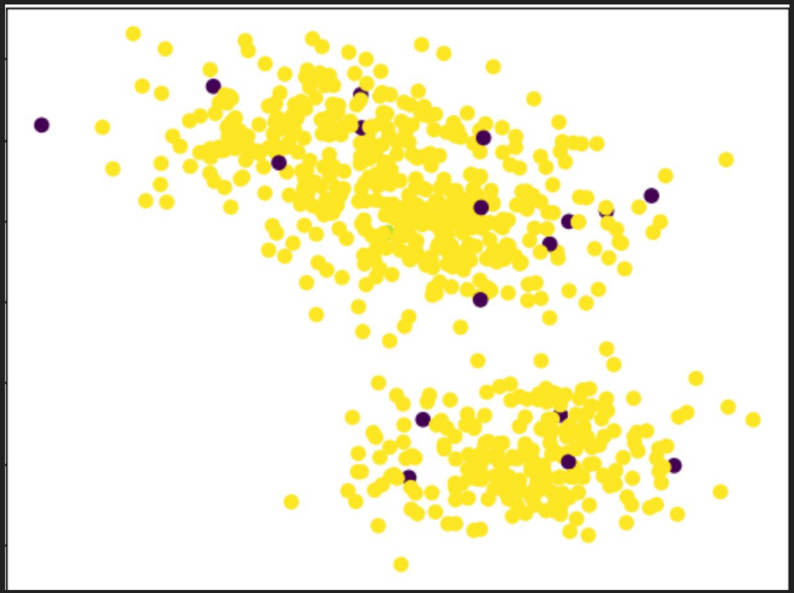
Results: Simulation Demonstration - 5 Weeks

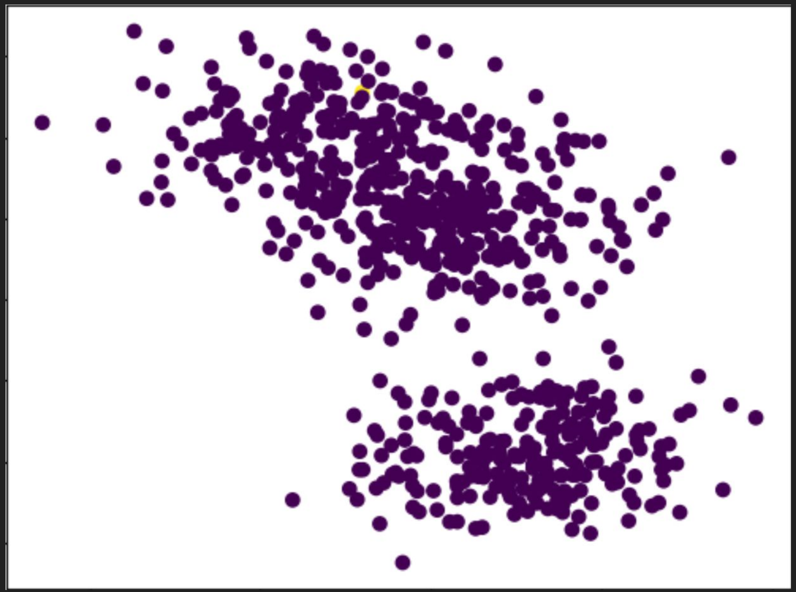




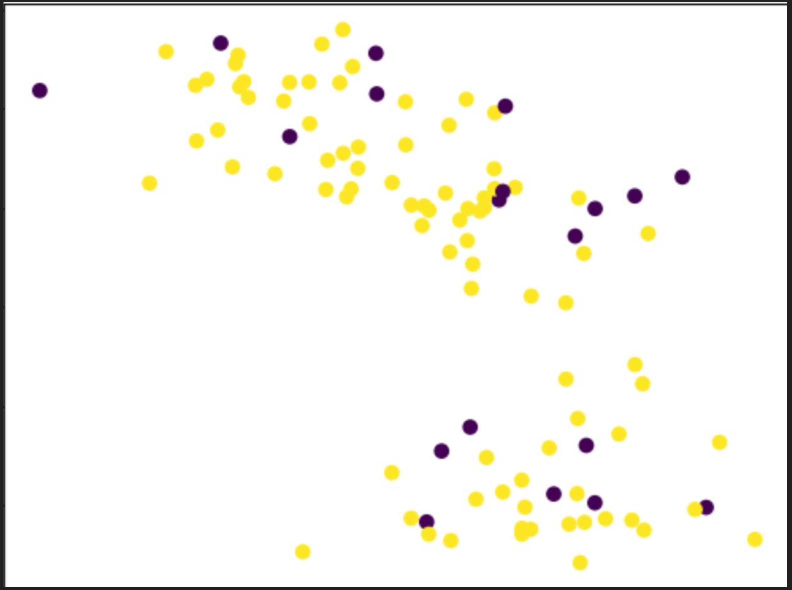


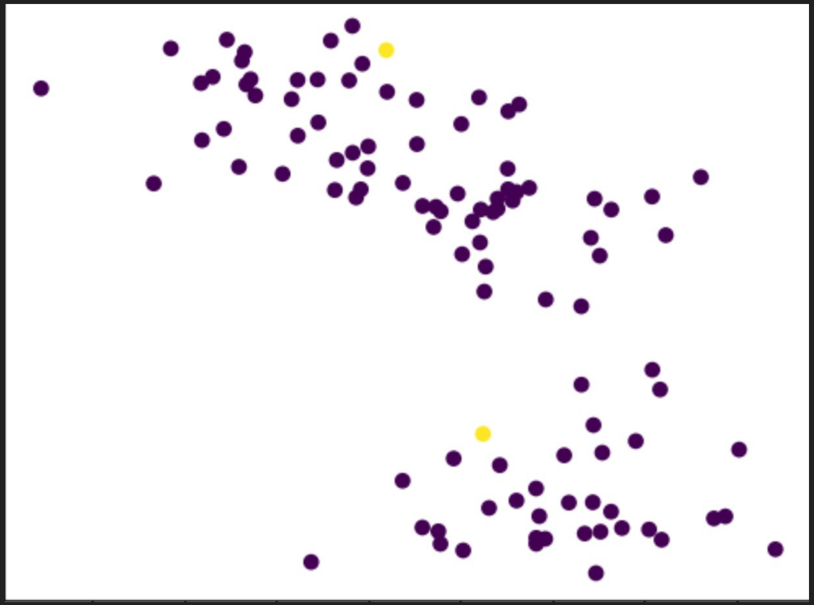
Results: Simulation Demonstration - 10 Weeks



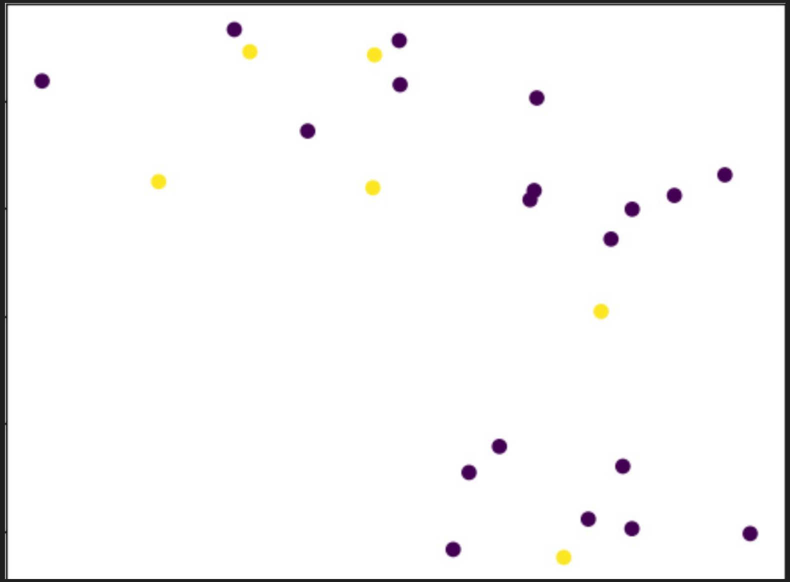


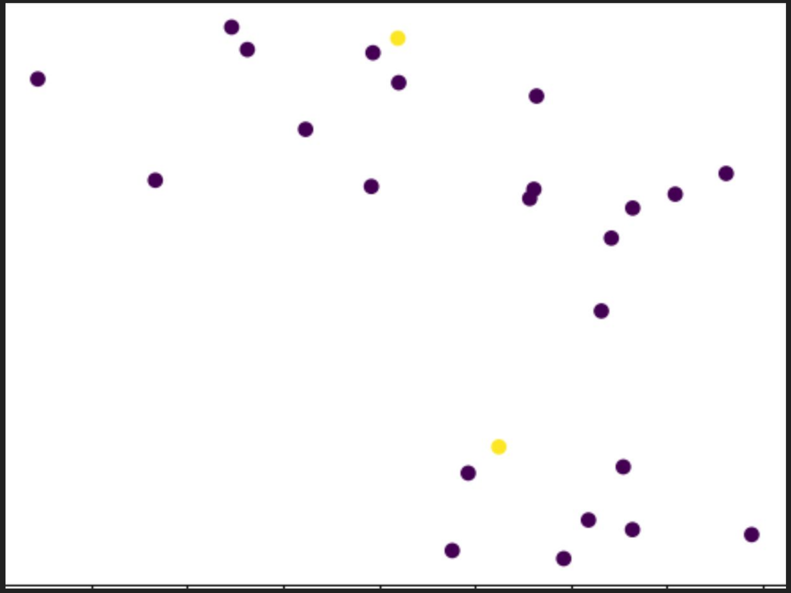
Results: Simulation Demonstration - 50 Weeks



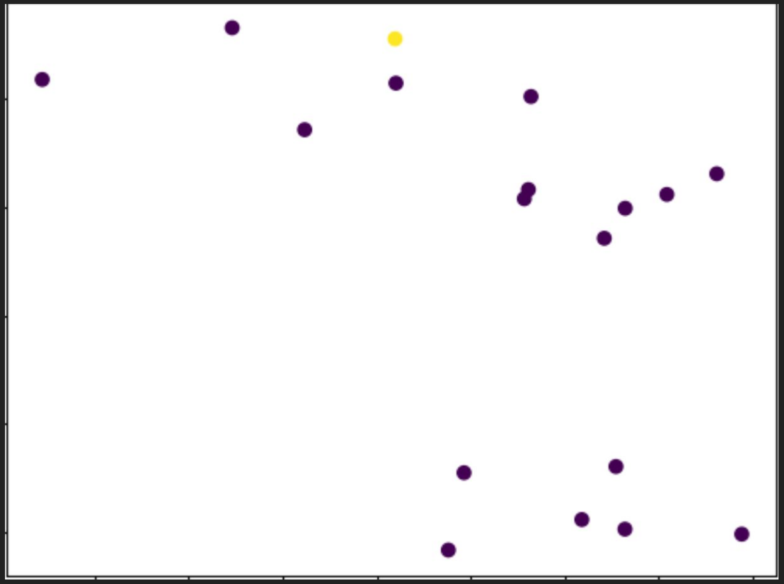


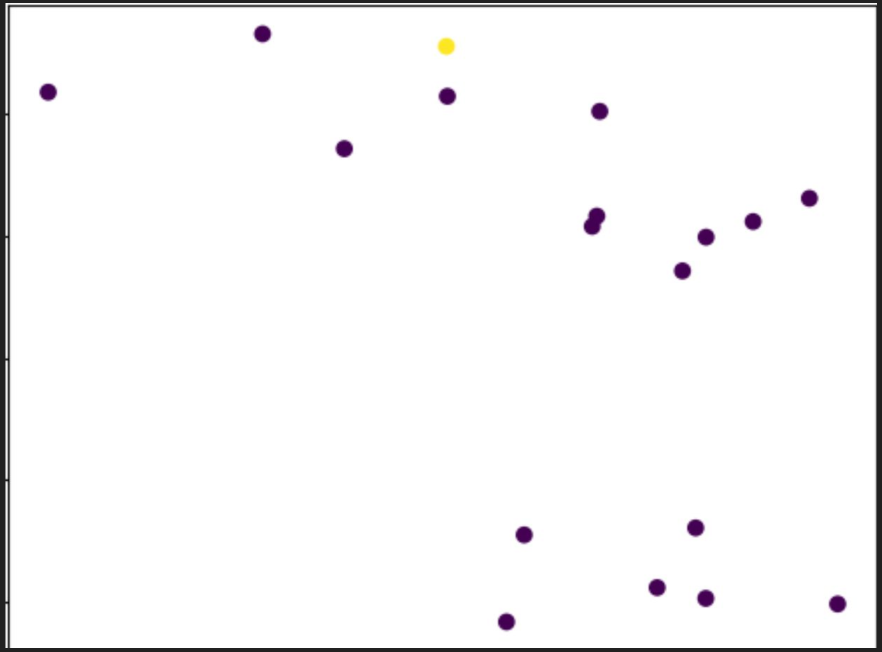
Results: Simulation Demonstration - 100 Weeks





Results: Simulation Demonstration - 500 Weeks

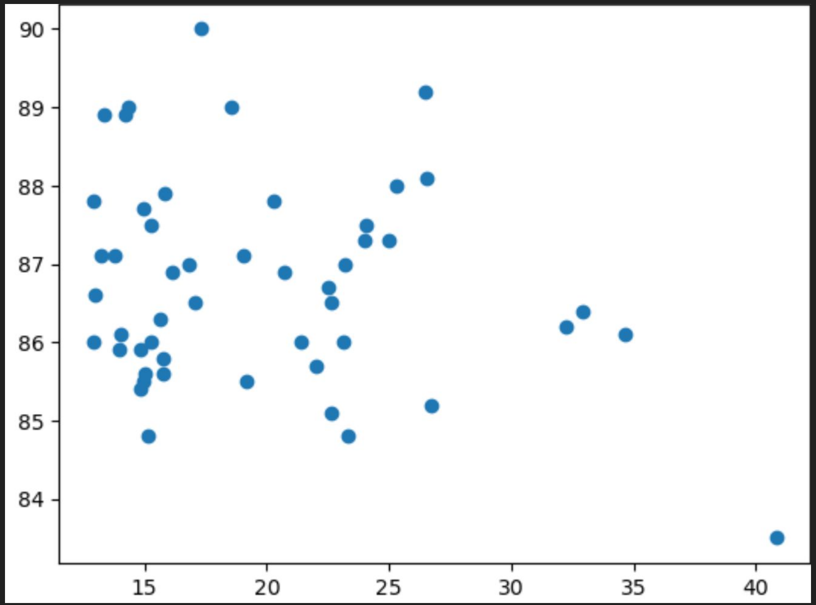




Results: Clinic Distance

Over a 100 simulations running for 100 weeks each, I compiled a plot of death rate in 100 weeks versus mean distance from clinic for each node, shown at right.

Casualty rate



Mean distance from clinic

Future Work: Response Strategies

Throughout the project I had a goal of making everything as flexible and expandable as possible, so I think there is a lot of room for future improvement to the project. The logic for deciding the placement of clinics is a changeable function that could be swapped from its current setting of mean distance minimization or random selection to a more intelligent (or perhaps learning based) strategy.

Future Work: Simulation Complexity

There is a lot of room for improved complexity of the simulation: data from my sources tracks distinctions in transmission rates and mortality rates based on demographic characteristics such as sex, class, and age that could be incorporated into the functions that decide tick behavior for each node in the graph.

Citations: Transmission and Demographics

- Hladik W, Stupp P, McCracken SD, Justman J, Ndongmo C, Shang J, Dokubo EK, Gummerson E, Koui I, Bodika S, Lobognon R, Brou H, Ryan C, Brown K, Nuwagaba-Biribonwoha H, Kingwara L, Young P, Bronson M, Chege D, Malewo O, Mengistu Y, Koen F, Jahn A, Auld A, Jonnalagadda S, Radin E, Hamunime N, Williams DB, Kayirangwa E, Mugisha V, Mdodo R, Delgado S, Kirungi W, Nelson L, West C, Biraro S, Dzekedzeke K, Barradas D, Mugurungi O, Balachandra S, Kilmarx PH, Musuka G, Patel H, Parekh B, Sleeman K, Domaoal RA, Rutherford G, Motsoane T, Bissek AZ, Farahani M, Voetsch AC. The epidemiology of HIV population viral load in twelve sub-Saharan African countries. PLoS One. 2023 Jun 26;18(6):e0275560. doi: 10.1371/journal.pone.0275560. PMID: 37363921; PMCID: PMC10292693.
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Thank you!

Questions?